

Ziyu Yao

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(Last update: May 16, 2026)

Research Interests

My research interests are mainly in three areas: (1) Machine Intelligence: advancing LLMs and LLM Agents in reasoning and planning (exemplar tasks include math reasoning, code generation, information extraction, and Web agent); (2) Responsible and Trustworthy Human-AI Interface: studying LLM interpretability (particularly Mechanistic Interpretability), human-AI interaction, and approaches facilitating the democratization of AI techniques; (3) Interdisciplinary Applications: bringing the impact of AI to domains such as Software Engineering and Education.

Education

- 2015–2021 **Ph.D.**, The Ohio State University (OSU), Computer Science & Engineering.
(Thesis: On Advancing Natural Language Interfaces: Data Collection, Model Development, and User Interaction. Advisor: Prof. Huan Sun)
- 2011–2015 **B.E.**, Beijing University of Posts and Telecommunications (BUPT), Communication Engineering.

Professional Experience

- 2021–present **Assistant Professor**, George Mason University (GMU).
- 2021 Summer **Research Intern**, Microsoft Corporation.
- 2020 Summer **Research Intern**, Carnegie Mellon University.
- 2018 Summer **Research Intern**, Microsoft Corporation.
- 2017 Summer **Research Intern**, Fujitsu Lab of America.
- 2014–2015 **Research Intern**, Tsinghua University, Beijing.

Awards and Honors

- 2026 **Amazon Faculty Award**.
- 2025 **Best Paper Honorable Mention**, [The First Workshop on the Application of LLM Explainability to Reasoning and Planning](#), COLM 2025.
- 2024 **Recognition for Service on Hangs-On Generative AI Tutorial**, GMU College of Engineering and Computing.
- 2023 **Top Reviewer**, NeurIPS.
- 2023 **Accelerating Foundation Models Research Program**, Microsoft Research.
- 2023 **Editor Commendation**, Journal of Business and Psychology.
- 2021 **Best Paper Award**, The IEEE International Conference on Bioinformatics and Biomedicine (BIBM).
- 2021 **Graduate Student Research Award**, OSU, CSE.
- 2020 **Presidential Fellowship**, OSU.
- 2020 **Rising Star in EECS**, UC Berkeley.
- 2020 **Graduate Students Poster Award (Honorable Mention)**, CSE, OSU.
- 2019 **NSF Student Travel Award**, The Web Conference (WWW).

- 2019 **Graduate Students Poster Award (Honorable Mention)**, CSE, OSU.
- 2018 **Travel Grant for *Graduate Cohort for Women*, San Francisco**, CRA-W.
- 2015 **Beijing Outstanding Graduate Award**, Beijing, China.
- 2012, 2014 **First-class Scholarship (top 3%)**, BUPT, China.
- 2013 **Second-class Scholarship (top 5%)**, BUPT, China.

Publications

(Notations: My Advisee; Myself; [†]Equal Contribution) .

PhD Dissertation.

Ziyu Yao, "On Advancing Natural Language Interfaces: Data Collection, Model Development, and User Interaction", The Ohio State University, PhD Dissertation, 2021. [[Dissertation](#)]

Preprints.

- 9 Rai, Daking, Mor Geva, and Ziyu Yao. "Data-driven Circuit Discovery for Interpretability of Language Models." arXiv preprint arXiv:2605.09129 (2026).
- 8 Yue, Murong, Zhiwei Liu, Liangwei Yang, Jianguo Zhang, Zuxin Liu, Haolin Chen, Ziyu Yao, Silvio Savarese, Caiming Xiong, Shelby Heinecke, and Huan Wang. "Toolibgen: Scalable automatic tool creation and aggregation for LLM reasoning." arXiv preprint arXiv:2510.07768 (2025).
- 7 Yan, Hao, Swapneel Suhas Vaidya, Xiaokuan Zhang, and Ziyu Yao. "Guiding AI to Fix Its Own Flaws: An Empirical Study on LLM-Driven Secure Code Generation." arXiv preprint arXiv:2506.23034 (2025).
- 6 Srivastava, Saurabh, and Ziyu Yao. "Revisiting Prompt Optimization with Large Reasoning Models-A Case Study on Event Extraction." arXiv preprint arXiv:2504.07357 (2025).
- 5 Mohamed Aghzal, Erion Plaku, Gregory J Stein, and Ziyu Yao, "A Survey on Large Language Models for Automated Planning", 2024. [[Paper](#)]
- 4 Daking Rai, Rydia R Weiland, Kayla Margaret Gabriella Herrera, Tyler H Shaw, and Ziyu Yao, "Understanding the Effect of Algorithm Transparency of Model Explanations in Text-to-SQL Semantic Parsing", 2024. [[Paper](#)]
- 3 Daking Rai, Yilun Zhou, Shi Feng, Abulhair Saparov, and Ziyu Yao, "A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models", 2024. [[Paper](#)][[GitHub Awesome-Mech-Interp-LM](#)][[X/Twitter](#)]
- 2 Hao Yan, Thomas D. Latoza, and Ziyu Yao, "IntelliExplain: Enhancing Conversational Code Generation for Non-Professional Programmers", 2024. [[Paper](#)][[Website](#)]
- 1 Qineng Wang, Chen Qian, Xiaochang Li, Ziyu Yao, and Huajie Shao, "Lens: A Foundation Model for Network Traffic", 2024. [[Paper](#)]

Accepted to Refereed Conference Proceedings.

- 34 Aghzal, Mohamed, Gregory J. Stein, and Ziyu Yao. "Why Do LLM-based Web Agents Fail? A Hierarchical Planning Perspective." Accepted to **ACL 2026**.
- 33 Yunxiang Peng, Mengmeng Ma, Ziyu Yao, and Xi Peng. "Inside-Out: Measuring Generalization in Vision Transformers Through Inner Working." Accepted to The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**).

- 32 Wenhan Lyu, Yimeng Wang, Murong Yue, Yifan Sun, Jennifer Suh, Meredith Kier, Ziyu Yao, and Yixuan Zhang. "Designing AI Peers for Collaborative Mathematical Problem Solving with Middle School Students: A Participatory Design Study." Accepted to **ACM CHI 2026**.
- 31 Rai, Daking, Samuel Miller, Kevin Moran, and Ziyu Yao. "Failure by Interference: Language Models Make Balanced Parentheses Errors When Faulty Mechanisms Overshadow Sound Ones." Accepted to **NeurIPS 2025**. **Won Best Paper Honorable Mention at XLLM-Reason-Plan Workshop at COLM 2025**
- 30 Siddarth Mamidanna, Daking Rai, Ziyu Yao, and Yilun Zhou. "All for One: LLMs Solve Mental Math at the Last Token With Information Transferred From Other Tokens." Accepted to **EMNLP 2025**. **Covered by AI influencer on X and "AI: post transformers" Podcast**
- 29 Zihao Li, Xu Wang, Yuzhe Yang, Ziyu Yao, Haoyi Xiong, and Mengnan Du. "Feature Extraction and Steering for Enhanced Chain-of-Thought Reasoning in Language Models." Accepted to **EMNLP 2025**. [[Paper](#)]
- 28 Dong Shu, Xuansheng Wu, Haiyan Zhao, Daking Rai, Ziyu Yao, Ninghao Liu, and Mengnan Du. "A survey on sparse autoencoders: Interpreting the internal mechanisms of large language models." Accepted to **EMNLP Findings 2025**. [[Paper](#)]
- 27 Pan, Jinhao, Chahat Raj, Ziyu Yao, and Ziwei Zhu. "Beneath the Surface: How Large Language Models Reflect Hidden Bias." Accepted to **EMNLP Findings 2025**. [[Paper](#)]
- 26 Murong Yue and Ziyu Yao. "Efficient but Vulnerable: Benchmarking and Defending LLM Batch Prompting Attack." Accepted to **Findings of ACL 2025**.
- 25 Saurabh Srivastava, Sweta Pati, and Ziyu Yao. "Instruction-Tuning LLMs for Event Extraction with Annotation Guidelines.; Accepted to **Findings of ACL 2025**.
- 24 Mohamed Aghzal, Xiang Yue, Erion Plaku, and Ziyu Yao, "Evaluating Vision-Language Models as Evaluators in Path Planning", To appear at **CVPR 2025**. [[Paper](#)]
- 23 Kong, Yangzhe, Daeun Song, Jing Liang, Dinesh Manocha, Ziyu Yao, and Xuesu Xiao. "AutoSpatial: Visual-Language Reasoning for Social Robot Navigation through Efficient Spatial Reasoning Learning." Accepted to **IROS 2025**.
- 22 Murong Yue, Wenlin Yao, Haitao Mi, Dian Yu, Ziyu Yao, and Dong Yu, "DOTS: Learning to Reason Dynamically in LLMs via Optimal Reasoning Trajectories Search", In *The Thirteenth International Conference on Learning Representations*. (**ICLR 2025**) [[Paper](#)] **Covered by MIT Tech Review China**
- 21 Yuqing Zhou, Ruixiang Tang, Ziyu Yao, and Ziwei Zhu, "Navigating the Shortcut Maze: A Comprehensive Analysis of Shortcut Learning in Text Classification by Language Models", In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2586–2614, Miami, Florida, USA. Association for Computational Linguistics. (**EMNLP Findings 2024**) [[Paper](#)][[Code](#)]
- 20 Mohamed Aghzal, Erion Plaku, and Ziyu Yao, "Look Further Ahead: Testing the Limits of GPT-4 in Path Planning." In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)*, pp. 1020-1027. IEEE, 2024. (**IEEE CASE 2024**) [[Paper](#)][[Code](#)]
- 19 Daking Rai and Ziyu Yao, "An Investigation of Neuron Activation as a Unified Lens to Explain Chain-of-Thought Eliciting Arithmetic Reasoning of LLMs", In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7174–7193, Bangkok, Thailand. Association for Computational Linguistics. (**ACL 2024**) [[Paper](#)][[Code](#)] **Covered by MIT Tech Review China (Chinese; English Translate)**

- 18 Saurabh Srivastava, Chengyue Huang, Weiguo Fan, and Ziyu Yao, “Instance Needs More Care: Rewriting Prompts for Instances Yields Better Zero-Shot Performance”, In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6211–6232, Bangkok, Thailand. Association for Computational Linguistics. (**ACL Findings 2024**) [[Paper](#)][[Code](#)]
- 17 Murong Yue, Jie Zhao, Min Zhang, and Liang Du, Ziyu Yao, “Large Language Model Cascades with Mixture of Thoughts Representations for Cost-efficient Reasoning”, In *The Twelfth International Conference on Learning Representations*. (**ICLR 2024**) [[Paper](#)][[Code](#)] [Featured in Hugging Face Daily Papers \(Oct 5, 2023\)](#) and [AWS Builder Blog](#)
- 16 Saurabh Srivastava, Gaurav Singh, Shou Matsumoto, and Ali Raz, Paulo Costa, Joshua Poore, Ziyu Yao, “MAILEX: Email Event and Argument Extraction”. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12964–12987, Singapore. Association for Computational Linguistics. (**EMNLP 2023**). [[Paper](#)][[Dataset&Code](#)]
- 15 Binfeng Xu, Xukun Liu, Hua Shen, and Zeyu Han, Yuhan Li, Murong Yue, Zhiyuan Peng, Yuchen Liu, Ziyu Yao, Dongkuan Xu, “Gentopia: A Collaborative Platform for Tool-Augmented LLMs”. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 237–245, Singapore. Association for Computational Linguistics. (**EMNLP 2023 Demo-Track**) [[Paper](#)][[Code](#)][[Project Website](#)]
- 14 Hao Yan, Saurabh Srivastava, Yintao Tai, Sida I. Wang, Scott Wen-tau Yih, Ziyu Yao, “Learning to Simulate Natural Language Feedback for Interactive Semantic Parsing”. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3149–3170, Toronto, Canada. Association for Computational Linguistics. (**ACL 2023**). [[Paper](#)][[Code](#)]
- 13 Daking Rai, Bailin Wang, Yilun Zhou, Ziyu Yao, “Improving Generalization in Language Model-based Text-to-SQL Semantic Parsing: Two Simple Semantic Boundary-based Techniques”. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 150–160, Toronto, Canada. Association for Computational Linguistics. (**ACL 2023**). [[Paper](#)][[Code](#)]
- 12 Daking Rai, Yilun Zhou, Bailin Wang, Ziyu Yao, “Explaining Large Language Model-based Neural Semantic Parsers” (Student Abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(13), 16308-16309. <https://doi.org/10.1609/aaai.v37i13.27014> (**AAAI 2023 Student Abstract**). [[Paper](#)]
- 11 Tianbao Xie[†], Chen Henry Wu[†], Peng Shi, Ruiqi Zhong, Torsten Scholak, Michihiro Yasunaga, Chien-Sheng Wu, Ming Zhong, Pengcheng Yin, Sida Wang, Victor Zhong, Bailin Wang, Chengzu Li, Connor Boyle, Ansong Ni, Ziyu Yao, Dragomir Radev, Caiming Xiong, Lingpeng Kong, Rui Zhang, Noah A. Smith, Luke Zettlemoyer, Tao Yu, “UnifiedSKG: Unifying and Multi-Tasking Structured Knowledge Grounding with Text-to-Text Language Models”. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 602–631, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics. (**EMNLP 2022, acceptance rate: 22.1%**). [[Paper](#), [Code](#), [Project Website](#)]
- 10 Xiang Yue, Ziyu Yao, Huan Sun, “Synthetic Question Value Estimation for Domain Adaptation of Question Answering”. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1340–1351, Dublin, Ireland. Association for Computational Linguistics. (**ACL 2022, acceptance rate: 20.75%**). [[Paper](#), [Code](#)]

- 9 Xiang Yue, Xinliang Frederick Zhang, Ziyu Yao, Simon Lin, Huan Sun, "CliniQG4QA: Generating Diverse Questions for Domain Adaptation of Clinical Question Answering". *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Houston, TX, USA, 2021, pp. 580-587, doi: 10.1109/BIBM52615.2021.9669300. **(IEEE BIBM 2021)** [[Paper](#), [Code](#)] **Best Paper Award**
- 8 Ziyu Yao, Frank Xu, Pengcheng Yin, Huan Sun, Graham Neubig, "Learning Structural Edits via Incremental Tree Transformations". In *The Ninth International Conference on Learning Representations (ICLR 2021)*. [[Paper](#), [Code](#)]
- 7 Ziyu Yao, Yiqi Tang, Scott Wen-tau Yih, Huan Sun, Yu Su, "An Imitation Game for Learning Semantic Parsers from User Interaction", In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6883–6902, Online. Association for Computational Linguistics. **(EMNLP 2020)**. [[Paper](#), [Code](#)]
- 6 Ziyu Yao, Yu Su, Huan Sun, Scott Wen-tau Yih, "Model-based Interactive Semantic Parsing: A Unified Formulation and A Text-to-SQL Case Study". In *2019 Conference on Empirical Methods in Natural Language Processing (EMNLP 2019)*. [[Paper](#), [Code](#)]
- 5 Boyuan Pan, Hao Li, Ziyu Yao, Deng Cai, Huan Sun, "Reinforced Dynamic Reasoning for Conversational Question Generation". In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019)*. [[Paper](#), [Code](#)]
- 4 Ziyu Yao, Jayavardhan Reddy Peddamail, Huan Sun, "CoaCor: Code Annotation for Code Retrieval with Reinforcement Learning", *The Web Conference 2019 (WWW 2019)*. [[Paper](#), [Code](#)]
- 3 Ziyu Yao, Xiujun Li, Jianfeng Gao, Brian Sadler, Huan Sun, "Interactive Semantic Parsing for If-Then Recipes via Hierarchical Reinforcement Learning", *The AAAI Conference on Artificial Intelligence 2019 (AAAI 2019)*. [[Paper](#), [Code](#)] **Spotlight**
- 2 Ziyu Yao, Daniel S. Weld, Wei-Peng Chen, Huan Sun, "StaQC: A Systematically Mined Question-Code Dataset from Stack Overflow", *The Web Conference 2018 (WWW 2018)*. [[Paper](#), [Code](#)]
- 1 Li Zhao, Minlie Huang, Ziyu Yao, Rongwei Su, Yingying Jiang, Xiaoyan Zhu. "Semi-Supervised Multinomial Naive Bayes for Text Classification by Leveraging Word-Level Statistical Constraint", *The AAAI Conference on Artificial Intelligence 2016 (AAAI 2016)*. [[Paper](#)]

Accepted to Refereed Workshop Proceedings.

- 7 Sarah Li, Ziyu Yao. "Evaluating the Effectiveness of Persona Simulation in Opinion Prediction with GPT-4.1." **IEEE ICDM 2025 Undergraduate and High School Symposium.**
- 6 Sai Adith Senthil Kumar, Hao Yan, Saipavan Perepa, Murong Yue, Ziyu Yao. "Can LLMs Simulate Personas with Reversed Performance? A Benchmark for Counterfactual Instruction Following." **Workshop on Social Simulation with LLMs at COLM, 2025.** [[Paper](#)]
- 5 Murong Yue[†], Wenhan Lyu[†], Wijdane Mifdal, Yixuan Zhang, Jennifer Suh, and Ziyu Yao, "MathVC: An LLM-Simulated Multi-Character Virtual Classroom for Mathematics Education", **AAAI AI4Edu Workshop, 2025.** [[Paper](#)][[Website](#)] **Covered by GovTech and GMU The George**
- 4 Sam Miller[†], Daking Rai[†], and Ziyu Yao, "Mechanistic Understanding of Language Models in Syntactic Code Completion", **AAAI Know4FM Workshop, 2025.** [[Paper](#)]

- 3 Mohamed Aghzal, Erion Plaku, and Ziyu Yao, "Can Large Language Models be Good Path Planners? A Benchmark and Investigation on Spatial-temporal Reasoning", **ICLR Workshop on LLM Agents, 2024**. [[Paper](#)][[Code](#)]
- 2 Peizhao Li, Xuchao Zhang, Ziyu Yao, Wei Cheng, Haifeng Chen, Hongfu Liu, "Code Editing from Few Exemplars by Adaptive Multi-Extent Composition", *The Deep Learning for Code Workshop at The Tenth International Conference on Learning Representations (ICLR 2022 DL4Code)*. [[Paper](#)]
- 1 Jayavardhan Reddy Peddamail, Ziyu Yao, Zhen Wang, Huan Sun, "A Comprehensive Study of StaQC for Deep Code Summarization", **SIGKDD 2018 Deep Learning Day**. [[Paper](#)]

Accepted to Refereed Journals.

- 3 Dipongkor, Atish Kumar, Ziyu Yao, and Kevin Moran. "Reassessing Code Authorship Attribution in the Era of Language Models." arXiv preprint arXiv:2506.17120 (2025). To Appear at ACM Transactions on Software Engineering and Methodology (TOSEM)
- 2 Philseok Lee, Shea Fyffe, Mina Son, Zihao Jia, and Ziyu Yao, "A paradigm shift from "human writing" to "machine generation" in personality test development: An application of state-of-the-art natural language processing", *Journal of Business and Psychology* (2023) 38:163–190, IF 6.604. [[Paper](#)] ([Editor Commendation, one of the 13 selected papers out of over 1,000 submissions in 2022](#))
- 1 Ziyu Yao, Shouzhong Tu, Minlie Huang, Xiaoyan Zhu. "A Semi-Supervised Method for Filtering Chinese Spam Tweets". *Journal of Chinese Information Processing* 2016, 30(5): 176-186. [[Paper in Chinese](#)]

Tutorials and Workshops

- 2025 **Tutorial on Mechanistic Interpretability for Language Models**, ICML 2025, Lead Speaker.
- 2025 **The First Workshop on the Application of LLM Explainability to Reasoning and Planning**, XLLM-Reason-Plan at COLM 2025, Lead Organizer.
- 2023 **10th Mid-Atlantic Student Colloquium on Speech, Language and Learning**, MASC-SLL 2023, Lead Organizer.
- 2022 **Workshop on Structured and Unstructured Knowledge Integration**, SUKI at NAACL 2022.
- 2021 **The First Workshop on NLP for Programming**, NLP4Prog at ACL 2021.

Invited Talks

- 2025 **From Science to Application of Language Models**, Invited talk at the Annual Summit of [VASEM \(Virginia Academy of Science, Engineering, and Medicine\)](#) representing GMU.
- 2025 **Mechanistically Interpreting Language Models: From Syntactic Code Completion to Arithmetic Reasoning.**, Invited talk at [DaSCI Seminar at the University of Granada](#). Host: Pablo Mesejo Santiago.
- 2025 **Interpretability Fills the Gap of Data Benchmarks.**, Invited talk at the BI4LLMC Workshop at FSE 2025. Host: Denys Poshyvanyk.
- 2025 **Language Model for Reasoning: The How and Why**, Invited talk at Georgetown University. Host: Nathan Schneider.
- 2024 **Mechanistic Understanding of LLMs in Reasoning and Code Generation**, Invited talk at University of Cambridge [[Publicity](#)].

- 2024 **MathVC: An LLM-Simulated Multi-Character Virtual Classroom for Mathematics Education**, Invited talk at [Wolfram Research, Online Colloquium for LLM Agents for Modeling Group Dynamics](#) [[Colloquium Website](#)].
- 2024 **The Progress, Opportunities, and Challenges of Semantic Parsing**, Invited talk at CTO Data Science Speaker Series at Bloomberg. Host: Daniel Preotiuc-Pietro [[Twitter/X Cover](#)].
- 2024 **Towards Understanding and Modeling the Interaction with Large Language Models**, Invited talk at [New York University](#). Host: Abulhair Saparov.
- 2024 **Interactive Semantic Parsing**, Invited talk at [Microsoft Research](#). Host: José Cambronero, Sumit Gulwani.
- 2024 **Building Natural Language Interfaces in the Age of Large Language Models**, Invited talk at Indiana University, Indianapolis. Host: Ming Jiang.
- 2024 **Towards Enhancing the Utilization of Large Language Models for Humans**, Invited talk at [Virginia Tech](#). Host: Chang-Tien Lu.
- 2024 **Towards Enhancing the Utilization of Large Language Models for Humans**, Invited talk at [University of Arizona](#). Host: Steven Bethard.
- 2023 **Building Natural Language Interfaces in the Age of Large Language Models**, Invited talk at [Tsinghua University](#). Host: Minlie Huang.
- 2023 **Large Language Model Cascades with Mixture of Thoughts Representations for Cost-efficient Reasoning**, Invited talk at [Microsoft Semantic Machines](#). Host: Hao Fang.
- 2023 **Building Natural Language Interfaces in the Age of Large Language Models**, Invited talk at [Google NLP Reading Group](#). Host: Pengcheng Yin.
- 2023 **Building Natural Language Interfaces in the Age of Large Language Models**, Invited talk at [Apple Inc.](#) Host: Yunyao Li.
- 2023 **Information Acquisition in the Age of Language Models**, Invited talk at [GMU C4I & Cyber Center](#)..
- 2022 **Challenges in Constructing Human-centric Natural Language Interfaces**, Invited talk at [GMU Statistics Department](#). Host: Nicholas Rios.
- 2022 **On Building Natural Language Interfaces for “Humans”**, Invited talk at [ServiceNow Research](#). Host: Dzmitry Bahdanau.
- 2022 **Building Interactive and Adaptive Question Answering Systems**, Invited talk at CLSP (Center for Language and Speech Processing) Seminar, Johns Hopkins University. Host: Kevin Duh, Ben Van Durme.
- 2022 **Building Interactive and Adaptive Question Answering Systems**, Invited talk at [GMU Center for Advancing Human-Machine Partnership](#)..
- 2021 **Building Interactive Natural Language Interfaces**, Invited talk at [Brandeis University](#). Host: Hongfu Liu.
- 2020 **On Building an Interactive Natural Language Interface: An intelligent agent that can communicate, collaborate and learn**, Invited talk at [Microsoft Research AI Breakthroughs Workshop](#) (virtually). Invited by: Ahmed Hassan Awadallah, Alex Polozov.
- 2020 **Building Natural Language Interfaces to Programs**, Invited talk at [VMware](#), Beijing (virtually). Host: Jin He, Kevin Song.
- 2020 **Towards Building Interactive and Collaborative Natural Language Interfaces**, Invited talk at [NeuLab, Carnegie Mellon University](#) (virtually). Host: Graham Neubig.

- 2019 **Towards Building Interactive and Collaborative Natural Language Interfaces**, Invited talk at the Chair of Management Information System, ETH Zurich, Host: Stefan Feuerriegel.

Research Funding

- 10 **"Agentic Mechanistic Interpretation of Language Models"**, Foresight Institute, \$100,000 (GMU: \$100,000, my part: \$100,000) with 10% overhead, 03/2026-05/2027. Role: PI.
- 9 **"PatriBot (Patriot GenAI Chatbot)"**, GMU Foundation, \$200,000 (my part: \$100,000) without overhead, 05/2024-05/2025. Role: Co-PI. (PI: Sarah Campbell at GMU)
- 8 **"Collaborative Research: Fostering Mathematical Modeling Competencies through Collaborative Learning in a Large Language Model (LLM) Simulated Virtual Classroom"**, National Science Foundation, \$899,920 (GMU: \$578,047, my part: ~\$300,000), 09/2024-08/2027. Role: PI (lead institution).
- 7 **"Securing the Supply Chain of LLMs as Software"**, Virginia Commonwealth Cyber Initiative, \$49,997 (GMU:\$49,997, my part: ~\$40,000) without overhead, 01/2024-06/2025. Role: PI.
- 6 **"A Foundation Model-Simulated Virtual Classroom for PreK-12 STEM Education"**, [Microsoft Research, Accelerating Foundation Models Research Program](#), \$30,000 Azure credits, 10/2023-04/2025. Role: PI.
- 5 **"Collaborative Research: SHF: Medium: Toward Understandability and Interpretability for Neural Language Models of Source Code"**, National Science Foundation, \$1,074,380 (GMU:\$745,197, my part: \$391,609), 10/2023-09/2027. Role: Co-PI → Site PI.
- 4 **"ARLIS Support for IARPA's STG AI-ML Solutions Program"**, Intelligence Advanced Research Projects Activity/UMD ARLIS, \$100,000 (GMU:\$100,000, my part: ~\$30,000), 10/2022-03/2024. Role: Co-PI. (PI: Ali Raz at GMU)
- 3 **"NSF Convergence Accelerator Track G: Secure Texting over Non-cooperative Networks and Anti-jamming Enhancement in 5G"**, National Science Foundation, 09/2022-07/2023. Role: Senior Personnel (1 summer month). (PI: Kai Zeng at GMU)
- 2 **"Securing Interactions Between Humans and Machines"**, Virginia Commonwealth Cyber Initiative, \$59,971 (GMU:\$59,971, my part: ~\$30,000) without overhead, 12/2022-12/2023. Role: PI.
- 1 **"AI Engineering Seedlings"**, US Department of Defense/UMD ARLIS, \$100,000 (GMU:\$100,000, my part: ~\$30,000), 07/2021-07/2022. Role: Co-PI. (PI: Ali Raz at GMU)

Teaching Experience

*denotes newly developed course

- 2025 Fall CS678: Advanced Natural Language Processing [\[Website\]](#).
- 2024 Fall CS478: Natural Language Processing [\[Website\]](#).
- 2024 Spring CS678: Advanced Natural Language Processing [\[Website\]](#).
- 2023 Fall CS678: Advanced Natural Language Processing [\[Website\]](#).
- 2023 Spring CS478*: Natural Language Processing [\[Website\]](#).
- 2022 Fall CS678*: Advanced Natural Language Processing [\[Website\]](#).
- 2022 Spring CS499: Natural Language Processing [\[Website\]](#).
- 2021 Fall CS695: Special Topics on Natural Language Processing [\[Website\]](#).

Mentoring Experience

Mentored PhD Students.

Daking Rai (Spring'22-)

Saurabh Srivastava (Spring'22-; **Outstanding GTA Award at GMU CS, 2025**)

Hao Yan (Spring'22-)

Murong Yue (Fall'22-; **Outstanding PhD Student at GMU CS, 2024**)

Mohamed Aghzal (Spring'23-)

Ayan Antik Khan (Fall'25-)

Mentored/Collaborated Master Students.

Sweta Pati (Summer'24-Spring'25)

Samuel Miller (Summer'24-Spring'25)

Gaurav Singh (Spring'22)

Janit Bidhan (Fall'22)

Mentored/Collaborated Undergraduate Students.

Siddarth Mamidanna (Spring'25-, remote intern from UCSC)

Sai Adith Senthil Kumar (Spring'24-, funded by VA CCI; **KEEN Award at GMU CEC Undergrad Research Celebration, 2025**)

Jiwon Chung (Spring'25)

Saipavan Perepa (Spring'24-Spring'25, funded by VA CCI)

Wijdane Mifdal (Fall'23-Spring'24, funded by OSCAR FW)

Qineng Wang (Spring'23-Winter'23, remote intern from ZJU → PhD@Northwestern)

Mariana Ritchie (Spring'22)

Brian Meike (Spring'22)

Soumithri Gadepalli (Spring'22)

Yintao Tai (Fall'21, remote intern from Univ. of Edinburgh)

Mentored/Collaborated High School Students.

Sarah Li (Summer'24 & '25, McLean High School, through ASSIP; **Third Place and the AFCEA Special Award at the Fairfax County Science Fair, 2025**)

Angela Xiang (Summer'24, Montgomery Blair High School, through ASSIP)

Mihir Kulshreshtha (Summer'23, Thomas Jefferson High School for Science and Technology, through ASSIP)

Anika Giri (Summer'23, Chantilly High School)

School/Department Services

Department Committees.

SWE Master Recruiting & TT Faculty Recruiting Committee (Fall'23 - Spring'24)

PhD Recruiting Committee (Fall'22 - Spring'23)

Faculty Search Committee (Fall'21 - Spring'22)

PhD Comprehensive Exam Committees.

Mehrdad Fazli (Fall'25)

Jonathan Hus (Fall'25)

Yuqing Zhou (Fall'25)

Rehab Alsudais (Fall'25)
Joshua Otten (Fall'24)
Nathaniel Krasner (Spring'24)
Murong Yue (Spring'24)
Saurabh Srivastava (Spring'24)
Daking Rai (Fall'23)
Hao Yan (Fall'23)
Syeda Sabrina Akter (Fall'23)
Milind Agarwal (Fall'23)
Fahim Faisal (Fall'22)
Jonathan Mbuya (Fall'22)
Joost Bottenbley (Fall'22)
Md Mahfuz Ibn Alam (Spring'22)
Abhisekh Rana (Spring'22)

PhD Thesis Proposal Committees.

Kourosh T Baghaei (Fall'25)
Md Nishat Raihan (Fall'25)
Nathaniel Krasner (Fall'25)
Prabin Bhandari (Spring'25)
Abhishek Paudel (Fall'24)
Milind Agarwal (Fall'24)
Jonathan Mbuya (Summer'24)
Fahim Faisal (Spring'24)
Kai North (Spring'24)
Anowarul Kabir (Spring'24)
Md Mahfuz Ibn Alam (Spring'23)

PhD/MS Thesis Defense Committees.

Abhishek Paudel (Fall'25)
Milind Agarwal (Fall'25)
Mamnuya Rinki (Fall'25, MS)
Jonathan Mbuya (Summer'25)
Kai North (Summer'24)
Md Mahfuz Ibn Alam (Summer'24)

Professional Activities

Outreach.

Faculty Mentor, CRA Undergraduate Research to PhD (UR2PhD)	2024
Invited Guest Lecturer at GMU School of Education	Spring 2024
Invited Guest Lecturer at GMU Department of Health Administration and Policy	Spring 2024
Invited Speaker at Morgan State University AI Camp	2023
Invited Speaker at "Girls Who Code" Chapter at Chantilly High School	2023
Panelist at NSF-funded Webinar about "Securely Operating Through 5G Infrastructure"	2023
Faculty Judge at the Science and Engineering Fair at Thomas Jefferson High School for Science and Technology	2023
Faculty Mentor at SIGKDD Undergraduate Consortium	2022

Professional Attendances.

NSF Workshop on the Science of Safe AI, UPenn (invite only) 2025
CMU SEI & NSF Workshop on U.S. Leadership in Software Engineering & AI Engineering: Critical Needs & Priorities, Alexandria, VA 2023

Organizing Committees.

[XLLM-Reason-Plan](#) at COLM'25: The First Workshop on the Application of LLM Explainability to Reasoning and Planning 2025
NAACL'24: 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics ([Diversity & Inclusion Chair](#)) 2024
[MASC-SLL](#): 10th Mid-Atlantic Student Colloquium on Speech, Language and Learning ([GMU's first-ever lead of this annual student event; we achieved a record number of 66 submissions and 130 attendees from ~30 organizations.](#)) 2023
[SUKI](#) at NAACL'22: Workshop on Structured and Unstructured Knowledge Integration 2022
[NLP4Prog](#) at ACL'21: The First Workshop on NLP for Programming 2021

Area Chairs or Senior Program Committees.

[ARR](#): ACL Rolling Review (This is an open review platform for papers submitted to ACL, EMNLP, NAACL, etc., starting from 2023) 2023 –
[ACL](#): Annual Meeting of the Association for Computational Linguistics 2023 –
[EMNLP](#): Conference on Empirical Methods in Natural Language Processing 2022 –
[NeurIPS](#): Annual Conference on Neural Information Processing Systems 2026 –
[ICLR](#): The International Conference on Learning Representations 2026 –
[AAAI](#): The AAAI Conference on Artificial Intelligence 2023,2024

Session Chair.

[AAAI](#): The AAAI Conference on Artificial Intelligence 2023

Program Committees/Reviewers (Conference).

[ARR](#): ACL Rolling Review 2021 –
[NeurIPS](#): Annual Conference on Neural Information Processing Systems 2023, 2025
[COLM](#): Conference on Language Modeling 2024, 2025
[ICLR](#): International Conference on Learning Representations 2024
[EACL SRW](#): EACL Student Research Workshop 2023
[NAACL](#): Annual Conference of the North American Chapter of the Association for Computational Linguistics 2021
[EMNLP](#): Conference on Empirical Methods in Natural Language Processing 2018, 2020, 2021
[MSR](#): Mining Software Repositories Conference 2021
[AACL](#): Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics 2020
[AAAI](#): The AAAI Conference on Artificial Intelligence 2020
[NLPCC](#): Conference on Natural Language Processing and Chinese Computing 2020
[CoNLL](#): Conference on Computational Natural Language Learning 2019

Secondary/External Reviewer:

[ACL](#): Annual Meeting of the Association for Computational Linguistics 2018, 2020
[KDD](#): SIGKDD Conference on Knowledge Discovery and Data Mining 2017, 2019, 2020
[NAACL](#): Annual Conference of the North American Chapter of the Association for Computational Linguistics 2019
[ICDM](#): International Conference on Data Mining 2016, 2018
[WWW](#): International World Wide Web Conference 2017
[CIKM](#): International Conference on Information and Knowledge Management 2016

Program Committees/Reviewers (Workshop).

<u>SRW at NAACL 2025</u> : NAACL Student Research Workshop	2025
<u>TRL at NeurIPS 2024</u> : 3th Table Representation Learning Workshop	2024
<u>NLP4PI at EMNLP 2024</u> : Workshop on NLP for Positive Impact	2024
<u>InterNLP at NeurIPS 2022</u> : Workshop on Interactive Learning for Natural Language Processing	2022
<u>BigData-IT at IEEE BigData 2022</u> : Workshop on Knowledge Discovery and Data Mining on IT Operations	2022
<u>HCI+NLP at NAACL 2022</u> : Bridging Human-Computer Interaction and Natural Language Processing	2022
<u>DL4Code at ICLR 2022</u> : Deep Learning for Code	2022
<u>IntEx-SemPar at EMNLP 2020</u> : Interactive and Executable Semantic Parsing Workshop	2020
<u>NLI at ACL 2020</u> : Natural Language Interface Workshop	2020
<u>SRW at ACL 2019</u> : Student Research Workshop	2019

Reviewers (Journal).

<u>Computational Linguistics</u> : Computational Linguistics, MIT Press	2024 –
<u>TACL</u> : Transactions of the Association for Computational Linguistics	2024 –
<u>TKDD</u> : ACM Transactions on Knowledge Discovery from Data	2022 –
<u>TKDE</u> : IEEE Transactions on Knowledge and Data Engineering	2022 –

Other Professional Services.

ACM SIGAI: AI Matters Newsletter Editor	2022 – 2025
National Science Foundation: Panel Reviewer	2023 –
<u>European Research Council (ERC)</u> : External Reviewer	2026